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In [1]: import os
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import EfficientNetB0
from tensorflow.keras.models import Model
from tensorflow.keras.layers import GlobalAveragePooling2D, Dense, Dropout
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.callbacks import EarlyStopping, ReduceLROnPlateau

from sklearn.metrics import classification_report, confusion_matrix
import seaborn as sns

# ignoring warnings
import warnings
warnings.simplefilter("ignore")
```

```
In [2]: # Parámetros generales
image_height = 224
image_width = 224
batch_size = 32
epochs = 20
input_shape = (image_height, image_width, 3)
num_classes = None # Se infiere del generador

data_dir = "Datos Finales"
metadata_file = "Metadatos_Finales.csv"

df = pd.read_csv(metadata_file, sep=";")
df = df.rename(columns={'Nombre del archivo': 'image_id'})
df['Planta_Estado_Español'] = df['Planta_Estado_Español'].astype(str)

categoria = 'Planta_Estado_Español'
image_id = 'image_id'
```

```
In [3]: # Dividir en entrenamiento y validación
from sklearn.model_selection import train_test_split

train_df, test_df = train_test_split(df, test_size=0.2, stratify=df['Planta_Estado_Español'],
train_df.shape, test_df.shape)
```

```
Out[3]: ((42354, 12), (10589, 12))
```

```
In [4]: train_df, val_df = train_test_split(train_df, test_size=0.2, stratify=train_df['Planta_Estado_Español'],
train_df.shape, val_df.shape)
```

Out[4]: ((33883, 12), (8471, 12))

```
In [5]: # Total de clases
print("Clases:", df['Planta_Estado_Español'].nunique())
num_clases = df['Planta_Estado_Español'].nunique()
class_names = df['Planta_Estado_Español'].unique().tolist()
print(class_names)
```

Clases: 18

['Yuca_Enferma', 'Yuca_Saludable', 'Fresa_Saludable', 'Papa_Enferma', 'Maíz_Saludable', 'Tomate_Enferma', 'Manzana_Enferma', 'Maíz_Enferma', 'Tomate_Saludable', 'Manzana_Saludable', 'Fresa_Enferma', 'Papa_Saludable', 'Coliflor_Enferma', 'Pepino Cohombro_Saludable', 'Fríjol_Enferma', 'Coliflor_Saludable', 'Fríjol_Saludable', 'Pepino Cohombro_Enferma']

```
In [6]: # Crear el generador de imágenes --
datagen = ImageDataGenerator(
    validation_split = 0.2,
    rescale=1./255,
    preprocessing_function = None,
    rotation_range = 45,
    zoom_range = 0.2,
    horizontal_flip = True,
    vertical_flip = True,
    fill_mode = 'nearest',
    shear_range = 0.1,
    height_shift_range = 0.1,
    width_shift_range = 0.1
)

train_generator = datagen.flow_from_dataframe(
    dataframe=train_df,
    directory=data_dir,
    x_col=image_id,
    y_col=categoria,
    target_size=(image_height, image_width),
    batch_size=batch_size,
    class_mode='categorical', # Importante para clasificación multiclase
    subset='training',
    seed=42,
    classes=class_names # Asegurarse del orden de las clases
)

validation_generator = datagen.flow_from_dataframe(
    dataframe=val_df,
    directory=data_dir,
    x_col=image_id,
    y_col=categoria,
    target_size=(image_height, image_width),
    batch_size=batch_size,
```

```

class_mode='categorical',
subset='validation',
seed=42,
classes=class_names
)

```

Found 27104 validated image filenames belonging to 18 classes.

Found 1694 validated image filenames belonging to 18 classes.

```

In [7]: base_model = EfficientNetB0(include_top=False, weights='imagenet', input_shape=(224, 224, 3))
base_model.trainable = False

x = base_model.output
x = GlobalAveragePooling2D()(x)
x = Dropout(0.3)(x)
output = Dense(num_classes, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=output)

model.compile(optimizer=Adam(learning_rate=1e-4), loss='categorical_crossentropy')
model.summary()

```

Model: "functional"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 224, 224, 3)	0	–
rescaling (Rescaling)	(None, 224, 224, 3)	0	input_layer[0]
normalization (Normalization)	(None, 224, 224, 3)	7	rescaling[0][0]
rescaling_1 (Rescaling)	(None, 224, 224, 3)	0	normalization
stem_conv_pad (ZeroPadding2D)	(None, 225, 225, 3)	0	rescaling_1[0]
stem_conv (Conv2D)	(None, 112, 112, 32)	864	stem_conv_pad
stem_bn (BatchNormalization)	(None, 112, 112, 32)	128	stem_conv[0][0]
stem_activation (Activation)	(None, 112, 112, 32)	0	stem_bn[0][0]
block1a_dwconv (DepthwiseConv2D)	(None, 112, 112, 32)	288	stem_activation

block1a_bn (BatchNormalizatio...	(None, 112, 112, 32)	128	block1a_dwcon
block1a_activation (Activation)	(None, 112, 112, 32)	0	block1a_bn[0]
block1a_se_squeeze (GlobalAveragePool...	(None, 32)	0	block1a_activ
block1a_se_reshape (Reshape)	(None, 1, 1, 32)	0	block1a_se_sq
block1a_se_reduce (Conv2D)	(None, 1, 1, 8)	264	block1a_se_re
block1a_se_expand (Conv2D)	(None, 1, 1, 32)	288	block1a_se_re
block1a_se_excite (Multiply)	(None, 112, 112, 32)	0	block1a_activ block1a_se_ex
block1a_project_co... (Conv2D)	(None, 112, 112, 16)	512	block1a_se_ex
block1a_project_bn (BatchNormalizatio...	(None, 112, 112, 16)	64	block1a_proje
block2a_expand_conv (Conv2D)	(None, 112, 112, 96)	1,536	block1a_proje
block2a_expand_bn (BatchNormalizatio...	(None, 112, 112, 96)	384	block2a_expan
block2a_expand_act... (Activation)	(None, 112, 112, 96)	0	block2a_expan
block2a_dwconv_pad (ZeroPadding2D)	(None, 113, 113, 96)	0	block2a_expan
block2a_dwconv (DepthwiseConv2D)	(None, 56, 56, 96)	864	block2a_dwcon
block2a_bn (BatchNormalizatio...	(None, 56, 56, 96)	384	block2a_dwcon
block2a_activation (Activation)	(None, 56, 56, 96)	0	block2a_bn[0]
block2a_se_squeeze (GlobalAveragePool...	(None, 96)	0	block2a_activ
block2a_se_reshape	(None, 1, 1, 96)	0	block2a_se_sq

(Reshape)			
block2a_se_reduce (Conv2D)	(None, 1, 1, 4)	388	block2a_se_re
block2a_se_expand (Conv2D)	(None, 1, 1, 96)	480	block2a_se_re
block2a_se_excite (Multiply)	(None, 56, 56, 96)	0	block2a_activ block2a_se_ex
block2a_project_co... (Conv2D)	(None, 56, 56, 24)	2,304	block2a_se_ex
block2a_project_bn (BatchNormalizatio...	(None, 56, 56, 24)	96	block2a_proje
block2b_expand_conv (Conv2D)	(None, 56, 56, 144)	3,456	block2a_proje
block2b_expand_bn (BatchNormalizatio...	(None, 56, 56, 144)	576	block2b_expan
block2b_expand_act... (Activation)	(None, 56, 56, 144)	0	block2b_expan
block2b_dwconv (DepthwiseConv2D)	(None, 56, 56, 144)	1,296	block2b_expan
block2b_bn (BatchNormalizatio...	(None, 56, 56, 144)	576	block2b_dwcon
block2b_activation (Activation)	(None, 56, 56, 144)	0	block2b_bn[0]
block2b_se_squeeze (GlobalAveragePool...	(None, 144)	0	block2b_activ
block2b_se_reshape (Reshape)	(None, 1, 1, 144)	0	block2b_se_sq
block2b_se_reduce (Conv2D)	(None, 1, 1, 6)	870	block2b_se_re
block2b_se_expand (Conv2D)	(None, 1, 1, 144)	1,008	block2b_se_re
block2b_se_excite (Multiply)	(None, 56, 56, 144)	0	block2b_activ block2b_se_ex
block2b_project_co... (Conv2D)	(None, 56, 56, 24)	3,456	block2b_se_ex

block2b_project_bn (BatchNormalizatio...	(None, 56, 56, 24)	96	block2b_proje
block2b_drop (Dropout)	(None, 56, 56, 24)	0	block2b_proje
block2b_add (Add)	(None, 56, 56, 24)	0	block2b_drop[block2a_proje
block3a_expand_conv (Conv2D)	(None, 56, 56, 144)	3,456	block2b_add[0
block3a_expand_bn (BatchNormalizatio...	(None, 56, 56, 144)	576	block3a_expan
block3a_expand_act...	(None, 56, 56, 144)	0	block3a_expan
block3a_dwconv_pad (ZeroPadding2D)	(None, 59, 59, 144)	0	block3a_expan
block3a_dwconv (DepthwiseConv2D)	(None, 28, 28, 144)	3,600	block3a_dwcon
block3a_bn (BatchNormalizatio...	(None, 28, 28, 144)	576	block3a_dwcon
block3a_activation (Activation)	(None, 28, 28, 144)	0	block3a_bn[0]
block3a_se_squeeze (GlobalAveragePool...	(None, 144)	0	block3a_activ
block3a_se_reshape (Reshape)	(None, 1, 1, 144)	0	block3a_se_sq
block3a_se_reduce (Conv2D)	(None, 1, 1, 6)	870	block3a_se_re
block3a_se_expand (Conv2D)	(None, 1, 1, 144)	1,008	block3a_se_re
block3a_se_excite (Multiply)	(None, 28, 28, 144)	0	block3a_activ block3a_se_ex
block3a_project_co...	(None, 28, 28, 40)	5,760	block3a_se_ex
block3a_project_bn (BatchNormalizatio...	(None, 28, 28, 40)	160	block3a_proje
block3b_expand_conv (Conv2D)	(None, 28, 28, 240)	9,600	block3a_proje

block3b_expand_bn (BatchNormalizatio...	(None, 28, 28, 240)	960	block3b_expan
block3b_expand_act...	(None, 28, 28, 240)	0	block3b_expan
block3b_dwconv (DepthwiseConv2D)	(None, 28, 28, 240)	6,000	block3b_expan
block3b_bn (BatchNormalizatio...	(None, 28, 28, 240)	960	block3b_dwcon
block3b_activation (Activation)	(None, 28, 28, 240)	0	block3b_bn[0]
block3b_se_squeeze (GlobalAveragePool...	(None, 240)	0	block3b_activ
block3b_se_reshape (Reshape)	(None, 1, 1, 240)	0	block3b_se_sq
block3b_se_reduce (Conv2D)	(None, 1, 1, 10)	2,410	block3b_se_re
block3b_se_expand (Conv2D)	(None, 1, 1, 240)	2,640	block3b_se_re
block3b_se_excite (Multiply)	(None, 28, 28, 240)	0	block3b_activ block3b_se_ex
block3b_project_co...	(None, 28, 28, 40)	9,600	block3b_se_ex
block3b_project_bn (BatchNormalizatio...	(None, 28, 28, 40)	160	block3b_proje
block3b_drop (Dropout)	(None, 28, 28, 40)	0	block3b_proje
block3b_add (Add)	(None, 28, 28, 40)	0	block3b_drop[block3a_proje
block4a_expand_conv (Conv2D)	(None, 28, 28, 240)	9,600	block3b_add[0
block4a_expand_bn (BatchNormalizatio...	(None, 28, 28, 240)	960	block4a_expan
block4a_expand_act...	(None, 28, 28, 240)	0	block4a_expan
block4a_dwconv_pad	(None, 29, 29,	0	block4a_expan

(ZeroPadding2D)	240)		
block4a_dwconv (DepthwiseConv2D)	(None, 14, 14, 240)	2,160	block4a_dwcon
block4a_bn (BatchNormalizatio...	(None, 14, 14, 240)	960	block4a_dwcon
block4a_activation (Activation)	(None, 14, 14, 240)	0	block4a_bn[0]
block4a_se_squeeze (GlobalAveragePool...	(None, 240)	0	block4a_activ
block4a_se_reshape (Reshape)	(None, 1, 1, 240)	0	block4a_se_sq
block4a_se_reduce (Conv2D)	(None, 1, 1, 10)	2,410	block4a_se_re
block4a_se_expand (Conv2D)	(None, 1, 1, 240)	2,640	block4a_se_re
block4a_se_excite (Multiply)	(None, 14, 14, 240)	0	block4a_activ block4a_se_ex
block4a_project_co... (Conv2D)	(None, 14, 14, 80)	19,200	block4a_se_ex
block4a_project_bn (BatchNormalizatio...	(None, 14, 14, 80)	320	block4a_proje
block4b_expand_conv (Conv2D)	(None, 14, 14, 480)	38,400	block4a_proje
block4b_expand_bn (BatchNormalizatio...	(None, 14, 14, 480)	1,920	block4b_expan
block4b_expand_act... (Activation)	(None, 14, 14, 480)	0	block4b_expan
block4b_dwconv (DepthwiseConv2D)	(None, 14, 14, 480)	4,320	block4b_expan
block4b_bn (BatchNormalizatio...	(None, 14, 14, 480)	1,920	block4b_dwcon
block4b_activation (Activation)	(None, 14, 14, 480)	0	block4b_bn[0]
block4b_se_squeeze (GlobalAveragePool...	(None, 480)	0	block4b_activ

block4b_se_reshape (Reshape)	(None, 1, 1, 480)	0	block4b_se_sq
block4b_se_reduce (Conv2D)	(None, 1, 1, 20)	9,620	block4b_se_re
block4b_se_expand (Conv2D)	(None, 1, 1, 480)	10,080	block4b_se_re
block4b_se_excite (Multiply)	(None, 14, 14, 480)	0	block4b_activ block4b_se_ex
block4b_project_co... (Conv2D)	(None, 14, 14, 80)	38,400	block4b_se_ex
block4b_project_bn (BatchNormalizatio...	(None, 14, 14, 80)	320	block4b_proje
block4b_drop (Dropout)	(None, 14, 14, 80)	0	block4b_proje
block4b_add (Add)	(None, 14, 14, 80)	0	block4b_drop[block4a_proje
block4c_expand_conv (Conv2D)	(None, 14, 14, 480)	38,400	block4b_add[0
block4c_expand_bn (BatchNormalizatio...	(None, 14, 14, 480)	1,920	block4c_expan
block4c_expand_act... (Activation)	(None, 14, 14, 480)	0	block4c_expan
block4c_dwconv (DepthwiseConv2D)	(None, 14, 14, 480)	4,320	block4c_expan
block4c_bn (BatchNormalizatio...	(None, 14, 14, 480)	1,920	block4c_dwcon
block4c_activation (Activation)	(None, 14, 14, 480)	0	block4c_bn[0]
block4c_se_squeeze (GlobalAveragePool...	(None, 480)	0	block4c_activ
block4c_se_reshape (Reshape)	(None, 1, 1, 480)	0	block4c_se_sq
block4c_se_reduce (Conv2D)	(None, 1, 1, 20)	9,620	block4c_se_re
block4c_se_expand (Conv2D)	(None, 1, 1, 480)	10,080	block4c_se_re

block4c_se_excite (Multiply)	(None, 14, 14, 480)	0	block4c_activ block4c_se_ex
block4c_project_co... (Conv2D)	(None, 14, 14, 80)	38,400	block4c_se_ex
block4c_project_bn (BatchNormalizatio...	(None, 14, 14, 80)	320	block4c_proje
block4c_drop (Dropout)	(None, 14, 14, 80)	0	block4c_proje
block4c_add (Add)	(None, 14, 14, 80)	0	block4c_drop[block4b_add[0
block5a_expand_conv (Conv2D)	(None, 14, 14, 480)	38,400	block4c_add[0
block5a_expand_bn (BatchNormalizatio...	(None, 14, 14, 480)	1,920	block5a_expan
block5a_expand_act... (Activation)	(None, 14, 14, 480)	0	block5a_expan
block5a_dwconv (DepthwiseConv2D)	(None, 14, 14, 480)	12,000	block5a_expan
block5a_bn (BatchNormalizatio...	(None, 14, 14, 480)	1,920	block5a_dwcon
block5a_activation (Activation)	(None, 14, 14, 480)	0	block5a_bn[0]
block5a_se_squeeze (GlobalAveragePool...	(None, 480)	0	block5a_activ
block5a_se_reshape (Reshape)	(None, 1, 1, 480)	0	block5a_se_sq
block5a_se_reduce (Conv2D)	(None, 1, 1, 20)	9,620	block5a_se_re
block5a_se_expand (Conv2D)	(None, 1, 1, 480)	10,080	block5a_se_re
block5a_se_excite (Multiply)	(None, 14, 14, 480)	0	block5a_activ block5a_se_ex
block5a_project_co... (Conv2D)	(None, 14, 14, 112)	53,760	block5a_se_ex
block5a_project_bn	(None, 14, 14,	448	block5a_proje

(BatchNormalizatio...	112)		
block5b_expand_conv (Conv2D)	(None, 14, 14, 672)	75,264	block5a_proje
block5b_expand_bn (BatchNormalizatio...	(None, 14, 14, 672)	2,688	block5b_expan
block5b_expand_act...	(None, 14, 14, 672)	0	block5b_expan
block5b_dwconv (DepthwiseConv2D)	(None, 14, 14, 672)	16,800	block5b_expan
block5b_bn (BatchNormalizatio...	(None, 14, 14, 672)	2,688	block5b_dwcon
block5b_activation (Activation)	(None, 14, 14, 672)	0	block5b_bn[0]
block5b_se_squeeze (GlobalAveragePool...	(None, 672)	0	block5b_activ
block5b_se_reshape (Reshape)	(None, 1, 1, 672)	0	block5b_se_sq
block5b_se_reduce (Conv2D)	(None, 1, 1, 28)	18,844	block5b_se_re
block5b_se_expand (Conv2D)	(None, 1, 1, 672)	19,488	block5b_se_re
block5b_se_excite (Multiply)	(None, 14, 14, 672)	0	block5b_activ block5b_se_ex
block5b_project_co...	(None, 14, 14, 112)	75,264	block5b_se_ex
block5b_project_bn (BatchNormalizatio...	(None, 14, 14, 112)	448	block5b_proje
block5b_drop (Dropout)	(None, 14, 14, 112)	0	block5b_proje
block5b_add (Add)	(None, 14, 14, 112)	0	block5b_drop[block5a_proje
block5c_expand_conv (Conv2D)	(None, 14, 14, 672)	75,264	block5b_add[0
block5c_expand_bn (BatchNormalizatio...	(None, 14, 14, 672)	2,688	block5c_expan

block5c_expand_act... (Activation)	(None, 14, 14, 672)	0	block5c_expan
block5c_dwconv (DepthwiseConv2D)	(None, 14, 14, 672)	16,800	block5c_expan
block5c_bn (BatchNormalizatio...	(None, 14, 14, 672)	2,688	block5c_dwcon
block5c_activation (Activation)	(None, 14, 14, 672)	0	block5c_bn[0]
block5c_se_squeeze (GlobalAveragePool...	(None, 672)	0	block5c_activ
block5c_se_reshape (Reshape)	(None, 1, 1, 672)	0	block5c_se_sq
block5c_se_reduce (Conv2D)	(None, 1, 1, 28)	18,844	block5c_se_re
block5c_se_expand (Conv2D)	(None, 1, 1, 672)	19,488	block5c_se_re
block5c_se_excite (Multiply)	(None, 14, 14, 672)	0	block5c_activ block5c_se_ex
block5c_project_co... (Conv2D)	(None, 14, 14, 112)	75,264	block5c_se_ex
block5c_project_bn (BatchNormalizatio...	(None, 14, 14, 112)	448	block5c_proje
block5c_drop (Dropout)	(None, 14, 14, 112)	0	block5c_proje
block5c_add (Add)	(None, 14, 14, 112)	0	block5c_drop[block5b_add[0
block6a_expand_conv (Conv2D)	(None, 14, 14, 672)	75,264	block5c_add[0
block6a_expand_bn (BatchNormalizatio...	(None, 14, 14, 672)	2,688	block6a_expan
block6a_expand_act... (Activation)	(None, 14, 14, 672)	0	block6a_expan
block6a_dwconv_pad (ZeroPadding2D)	(None, 17, 17, 672)	0	block6a_expan
block6a_dwconv (DepthwiseConv2D)	(None, 7, 7, 672)	16,800	block6a_dwcon

block6a_bn (BatchNormalizatio...	(None, 7, 7, 672)	2,688	block6a_dwcon
block6a_activation (Activation)	(None, 7, 7, 672)	0	block6a_bn[0]
block6a_se_squeeze (GlobalAveragePool...	(None, 672)	0	block6a_activ
block6a_se_reshape (Reshape)	(None, 1, 1, 672)	0	block6a_se_sq
block6a_se_reduce (Conv2D)	(None, 1, 1, 28)	18,844	block6a_se_re
block6a_se_expand (Conv2D)	(None, 1, 1, 672)	19,488	block6a_se_re
block6a_se_excite (Multiply)	(None, 7, 7, 672)	0	block6a_activ block6a_se_ex
block6a_project_co... (Conv2D)	(None, 7, 7, 192)	129,024	block6a_se_ex
block6a_project_bn (BatchNormalizatio...	(None, 7, 7, 192)	768	block6a_proje
block6b_expand_conv (Conv2D)	(None, 7, 7, 1152)	221,184	block6a_proje
block6b_expand_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block6b_expan
block6b_expand_act... (Activation)	(None, 7, 7, 1152)	0	block6b_expan
block6b_dwconv (DepthwiseConv2D)	(None, 7, 7, 1152)	28,800	block6b_expan
block6b_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block6b_dwcon
block6b_activation (Activation)	(None, 7, 7, 1152)	0	block6b_bn[0]
block6b_se_squeeze (GlobalAveragePool...	(None, 1152)	0	block6b_activ
block6b_se_reshape (Reshape)	(None, 1, 1, 1152)	0	block6b_se_sq
block6b_se_reduce	(None, 1, 1, 48)	55,344	block6b_se_re

(Conv2D)			
block6b_se_expand (Conv2D)	(None, 1, 1, 1152)	56,448	block6b_se_re
block6b_se_excite (Multiply)	(None, 7, 7, 1152)	0	block6b_activ block6b_se_ex
block6b_project_co... (Conv2D)	(None, 7, 7, 192)	221,184	block6b_se_ex
block6b_project_bn (BatchNormalizatio...	(None, 7, 7, 192)	768	block6b_proje
block6b_drop (Dropout)	(None, 7, 7, 192)	0	block6b_proje
block6b_add (Add)	(None, 7, 7, 192)	0	block6b_drop[block6a_proje
block6c_expand_conv (Conv2D)	(None, 7, 7, 1152)	221,184	block6b_add[0
block6c_expand_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block6c_expan
block6c_expand_act... (Activation)	(None, 7, 7, 1152)	0	block6c_expan
block6c_dwconv (DepthwiseConv2D)	(None, 7, 7, 1152)	28,800	block6c_expan
block6c_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block6c_dwcon
block6c_activation (Activation)	(None, 7, 7, 1152)	0	block6c_bn[0]
block6c_se_squeeze (GlobalAveragePool...	(None, 1152)	0	block6c_activ
block6c_se_reshape (Reshape)	(None, 1, 1, 1152)	0	block6c_se_sq
block6c_se_reduce (Conv2D)	(None, 1, 1, 48)	55,344	block6c_se_re
block6c_se_expand (Conv2D)	(None, 1, 1, 1152)	56,448	block6c_se_re
block6c_se_excite (Multiply)	(None, 7, 7, 1152)	0	block6c_activ block6c_se_ex

block6c_project_co... (Conv2D)	(None, 7, 7, 192)	221,184	block6c_se_ex
block6c_project_bn (BatchNormalizatio...	(None, 7, 7, 192)	768	block6c_proje
block6c_drop (Dropout)	(None, 7, 7, 192)	0	block6c_proje
block6c_add (Add)	(None, 7, 7, 192)	0	block6c_drop[block6b_add[0
block6d_expand_conv (Conv2D)	(None, 7, 7, 1152)	221,184	block6c_add[0
block6d_expand_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block6d_expan
block6d_expand_act... (Activation)	(None, 7, 7, 1152)	0	block6d_expan
block6d_dwconv (DepthwiseConv2D)	(None, 7, 7, 1152)	28,800	block6d_expan
block6d_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block6d_dwcon
block6d_activation (Activation)	(None, 7, 7, 1152)	0	block6d_bn[0]
block6d_se_squeeze (GlobalAveragePool...	(None, 1152)	0	block6d_activ
block6d_se_reshape (Reshape)	(None, 1, 1, 1152)	0	block6d_se_sq
block6d_se_reduce (Conv2D)	(None, 1, 1, 48)	55,344	block6d_se_re
block6d_se_expand (Conv2D)	(None, 1, 1, 1152)	56,448	block6d_se_re
block6d_se_excite (Multiply)	(None, 7, 7, 1152)	0	block6d_activ block6d_se_ex
block6d_project_co... (Conv2D)	(None, 7, 7, 192)	221,184	block6d_se_ex
block6d_project_bn (BatchNormalizatio...	(None, 7, 7, 192)	768	block6d_proje
block6d_drop (Dropout)	(None, 7, 7, 192)	0	block6d_proje

block6d_add (Add)	(None, 7, 7, 192)	0	block6d_drop[block6c_add[0
block7a_expand_conv (Conv2D)	(None, 7, 7, 1152)	221,184	block6d_add[0
block7a_expand_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block7a_expan
block7a_expand_act... (Activation)	(None, 7, 7, 1152)	0	block7a_expan
block7a_dwconv (DepthwiseConv2D)	(None, 7, 7, 1152)	10,368	block7a_expan
block7a_bn (BatchNormalizatio...	(None, 7, 7, 1152)	4,608	block7a_dwcon
block7a_activation (Activation)	(None, 7, 7, 1152)	0	block7a_bn[0]
block7a_se_squeeze (GlobalAveragePool...	(None, 1152)	0	block7a_activ
block7a_se_reshape (Reshape)	(None, 1, 1, 1152)	0	block7a_se_sq
block7a_se_reduce (Conv2D)	(None, 1, 1, 48)	55,344	block7a_se_re
block7a_se_expand (Conv2D)	(None, 1, 1, 1152)	56,448	block7a_se_re
block7a_se_excite (Multiply)	(None, 7, 7, 1152)	0	block7a_activ block7a_se_ex
block7a_project_co... (Conv2D)	(None, 7, 7, 320)	368,640	block7a_se_ex
block7a_project_bn (BatchNormalizatio...	(None, 7, 7, 320)	1,280	block7a_proje
top_conv (Conv2D)	(None, 7, 7, 1280)	409,600	block7a_proje
top_bn (BatchNormalizatio...	(None, 7, 7, 1280)	5,120	top_conv[0][0
top_activation (Activation)	(None, 7, 7, 1280)	0	top_bn[0][0]
global_average_poo...	(None, 1280)	0	top_activatio

(GlobalAveragePool...			
dropout (Dropout)	(None, 1280)	0	global_averag
dense (Dense)	(None, 18)	23,058	dropout[0][0]

Total params: 4,072,629 (15.54 MB)

Trainable params: 23,058 (90.07 KB)

Non-trainable params: 4,049,571 (15.45 MB)

```
In [8]: early_stop = EarlyStopping(monitor='val_loss', patience=5, restore_best_weights=True)
        reduce_lr = ReduceLROnPlateau(monitor='val_loss', factor=0.2, patience=3, verbose=1)
```

```
In [9]: history = model.fit(
        train_generator,
        steps_per_epoch=train_generator.samples // train_generator.batch_size,
        validation_data=validation_generator,
        validation_steps=validation_generator.samples // validation_generator.batch_size,
        epochs=epochs,
        callbacks=[early_stop, reduce_lr]
    )
```

Epoch 1/20

847/847 ————— **394s** 462ms/step - accuracy: 0.3192 - loss: 2.1073 - val_accuracy: 0.5337 - val_loss: 1.9264 - learning_rate: 1.0000e-04

Epoch 2/20

847/847 ————— **414s** 488ms/step - accuracy: 0.3443 - loss: 1.9312 - val_accuracy: 0.3456 - val_loss: 1.9126 - learning_rate: 1.0000e-04

Epoch 3/20

847/847 ————— **428s** 505ms/step - accuracy: 0.3581 - loss: 1.9427 - val_accuracy: 0.3407 - val_loss: 1.9115 - learning_rate: 1.0000e-04

Epoch 4/20

847/847 ————— **430s** 507ms/step - accuracy: 0.3662 - loss: 1.9259 - val_accuracy: 0.3425 - val_loss: 1.9080 - learning_rate: 1.0000e-04

Epoch 5/20

847/847 ————— **437s** 516ms/step - accuracy: 0.3666 - loss: 1.9371 - val_accuracy: 0.3419 - val_loss: 1.9114 - learning_rate: 1.0000e-04

Epoch 6/20

847/847 ————— **436s** 514ms/step - accuracy: 0.3709 - loss: 1.9318 - val_accuracy: 0.4243 - val_loss: 1.9041 - learning_rate: 1.0000e-04

Epoch 7/20

847/847 ————— **436s** 515ms/step - accuracy: 0.3763 - loss: 1.9325 - val_accuracy: 0.5793 - val_loss: 1.9063 - learning_rate: 1.0000e-04

Epoch 8/20

847/847 ————— **440s** 520ms/step - accuracy: 0.3792 - loss: 1.9340 - val_accuracy: 0.5974 - val_loss: 1.9010 - learning_rate: 1.0000e-04

Epoch 9/20

847/847 ————— **443s** 522ms/step - accuracy: 0.3829 - loss: 1.9172 - val_accuracy: 0.3492 - val_loss: 1.9065 - learning_rate: 1.0000e-04

Epoch 10/20
847/847 ————— **442s** 521ms/step - accuracy: 0.3893 - loss: 1.9272 - val_accuracy: 0.3431 - val_loss: 1.8985 - learning_rate: 1.0000e-04
 Epoch 11/20
847/847 ————— **440s** 519ms/step - accuracy: 0.3905 - loss: 1.9243 - val_accuracy: 0.3401 - val_loss: 1.9053 - learning_rate: 1.0000e-04
 Epoch 12/20
847/847 ————— **447s** 528ms/step - accuracy: 0.3937 - loss: 1.9102 - val_accuracy: 0.5715 - val_loss: 1.8966 - learning_rate: 1.0000e-04
 Epoch 13/20
847/847 ————— **443s** 523ms/step - accuracy: 0.3921 - loss: 1.9204 - val_accuracy: 0.3504 - val_loss: 1.8899 - learning_rate: 1.0000e-04
 Epoch 14/20
847/847 ————— **444s** 524ms/step - accuracy: 0.3961 - loss: 1.9193 - val_accuracy: 0.5938 - val_loss: 1.9035 - learning_rate: 1.0000e-04
 Epoch 15/20
847/847 ————— **3400s** 4s/step - accuracy: 0.3996 - loss: 1.9137 - val_accuracy: 0.3425 - val_loss: 1.8942 - learning_rate: 1.0000e-04
 Epoch 16/20
847/847 ————— **0s** 436ms/step - accuracy: 0.4041 - loss: 1.9179
 Epoch 16: ReduceLRonPlateau reducing learning rate to 1.9999999494757503e-05
847/847 ————— **393s** 464ms/step - accuracy: 0.4041 - loss: 1.9179 - val_accuracy: 0.3546 - val_loss: 1.8965 - learning_rate: 1.0000e-04
 Epoch 17/20
847/847 ————— **409s** 483ms/step - accuracy: 0.3995 - loss: 1.9104 - val_accuracy: 0.5126 - val_loss: 1.8932 - learning_rate: 2.0000e-05
 Epoch 18/20
847/847 ————— **415s** 490ms/step - accuracy: 0.4075 - loss: 1.9028 - val_accuracy: 0.5595 - val_loss: 1.8940 - learning_rate: 2.0000e-05

```
In [10]: print("\nEvaluando el modelo...")
evaluation_result = model.evaluate(validation_generator)
print(f"Pérdida en la validación: {evaluation_result[0]:.4f}")
print(f"Precisión en la validación: {evaluation_result[1]:.4f}")
```

Evaluando el modelo...

53/53 ————— **24s** 455ms/step - accuracy: 0.3295 - loss: 1.9669
 Pérdida en la validación: 1.8986
 Precisión en la validación: 0.3477

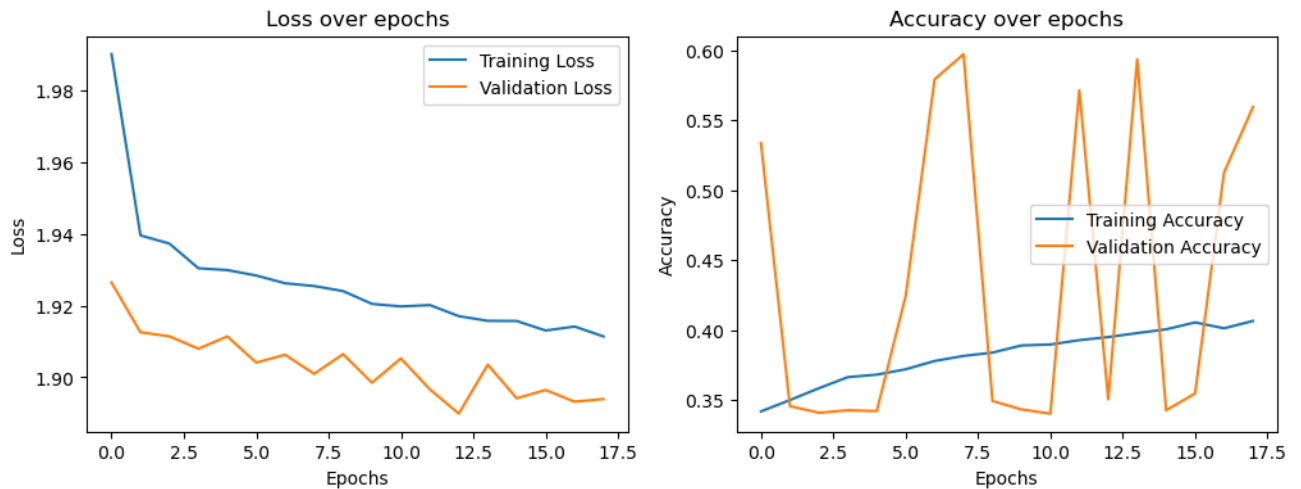
```
In [11]: # Acceder a los datos de pérdida y precisión
loss = history.history['loss']
val_loss = history.history['val_loss']
accuracy = history.history['accuracy']
val_accuracy = history.history['val_accuracy']

# Graficar la pérdida del modelo
plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
```

```
plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Loss over epochs')
plt.xlabel('Epochs')
plt.ylabel('Loss')
plt.legend()

# Graficar la precisión del modelo
plt.subplot(1, 2, 2)
plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Accuracy over epochs')
plt.xlabel('Epochs')
plt.ylabel('Accuracy')
plt.legend()
```

Out[11]: <matplotlib.legend.Legend at 0x1482ae600>



```
In [12]: epochs = list(range(1, 20))

loss = history.history['loss']
val_loss = history.history['val_loss']
accuracy = history.history['accuracy']
val_accuracy = history.history['val_accuracy']

data = {
    'Epoch': 20,
    'Training Loss': loss,
    'Validation Loss': val_loss,
    'Training Accuracy': accuracy,
    'Validation Accuracy': val_accuracy
}
df = pd.DataFrame(data)

# Imprimir la tabla
```

```
print(df)
```

	Epoch	Trainning Loss	Validation Loss	Training Accuracy \
0	20	1.990156	1.926441	0.341721
1	20	1.939595	1.912570	0.349838
2	20	1.937304	1.911452	0.358287
3	20	1.930435	1.907983	0.366256
4	20	1.929922	1.911417	0.368027
5	20	1.928405	1.904125	0.371864
6	20	1.926232	1.906312	0.377804
7	20	1.925472	1.900975	0.381420
8	20	1.924067	1.906473	0.383781
9	20	1.920490	1.898522	0.388909
10	20	1.919784	1.905284	0.389610
11	20	1.920154	1.896644	0.392710
12	20	1.917067	1.889883	0.394923
13	20	1.915760	1.903545	0.397764
14	20	1.915724	1.894160	0.400458
15	20	1.913065	1.896485	0.405401
16	20	1.914172	1.893216	0.401195
17	20	1.911406	1.893958	0.406434

	Validation Accuracy
0	0.533654
1	0.345553
2	0.340745
3	0.342548
4	0.341947
5	0.424279
6	0.579327
7	0.597356
8	0.349159
9	0.343149
10	0.340144
11	0.571514
12	0.350361
13	0.593750
14	0.342548
15	0.354567
16	0.512620
17	0.559495

```
In [13]: # Accuracy final en entrenamiento
val_loss = history.history['val_loss'][-1]
```

```
In [14]: print('Pérdida final en Train:', history.history['loss'][-1])
print("Pérdida final en Validation:", history.history['val_loss'][-1])

print("Accuracy final en Train:", history.history['accuracy'][-1])
```

```
print("Accuracy final en Validation:", history.history['val_accuracy'][-1])
```

Perdida final en Train: 1.9114060401916504
 Pérdida final en Validation: 1.8939577341079712
 Accuracy final en Train: 0.40643447637557983
 Accuracy final en Validation: 0.559495210647583

```
In [18]: test_df.Planta_Estado_Español = test_df.Planta_Estado_Español.astype('str')

test_generator = datagen.flow_from_dataframe(test_df,
                                             directory = data_dir,
                                             x_col = image_id,
                                             y_col = categoria,
                                             target_size = (image_height, image_width),
                                             batch_size = batch_size,
                                             class_mode = "categorical")
```

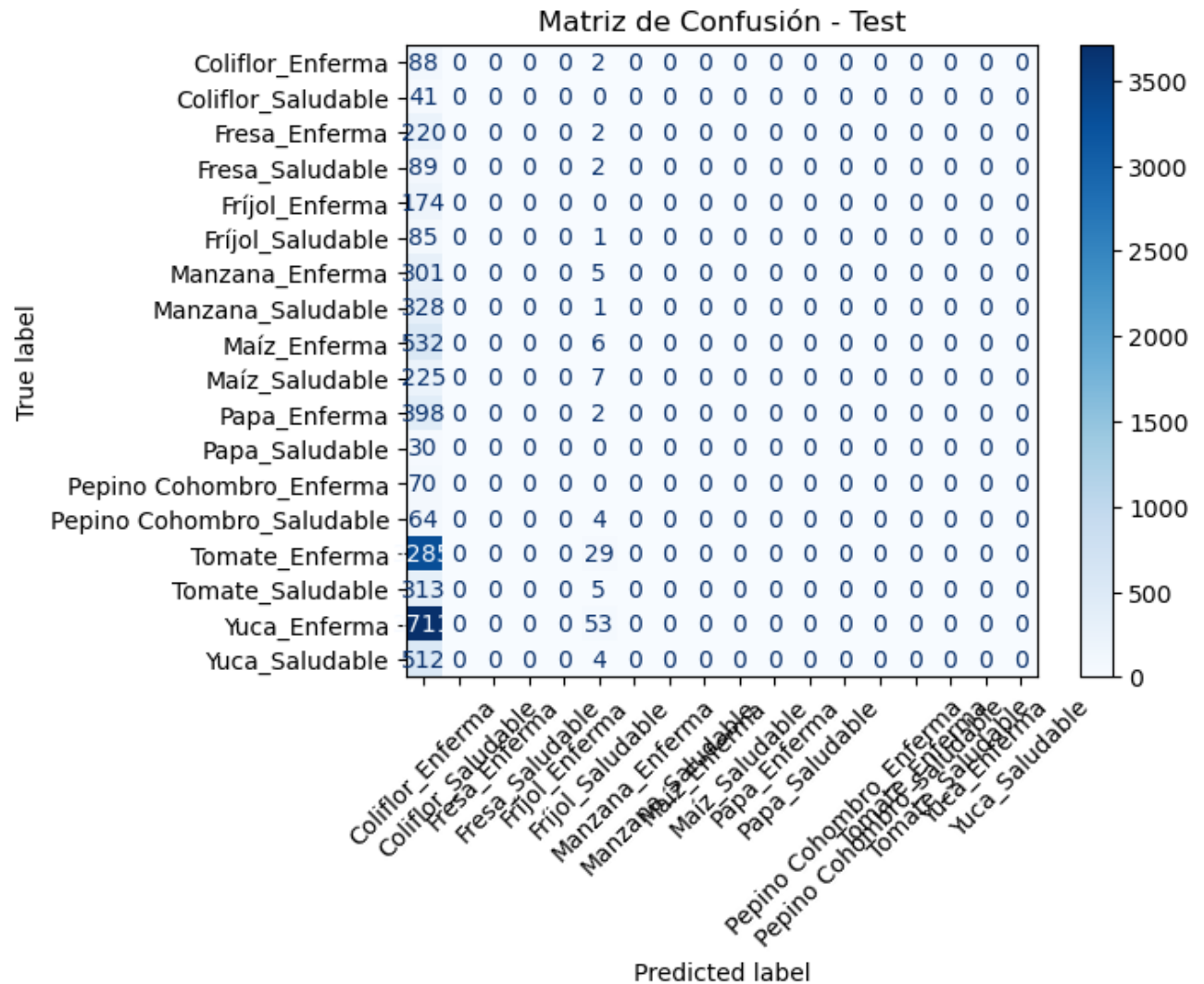
Found 10589 validated image filenames belonging to 18 classes.

```
In [19]: # -----
# 1. Obtener predicciones completas para test
# -----
y_pred_probs = model.predict(test_generator)
y_pred = np.argmax(y_pred_probs, axis=1)
y_true = test_generator.classes
```

331/331 ————— 141s 422ms/step

```
In [22]: # -----
# 2. Matriz de Confusión
# -----
from sklearn.metrics import confusion_matrix, accuracy_score, ConfusionMatrixDisplay

cm = confusion_matrix(y_true, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=test_generator.classes)
disp.plot(cmap="Blues", xticks_rotation=45)
plt.title("Matriz de Confusión - Test")
plt.show()
```



```
In [23]: from sklearn.metrics import confusion_matrix, accuracy_score, ConfusionMatrix
accuracy = accuracy_score(y_true, y_pred)
print(f"Test Accuracy: {accuracy:.4f}")
```

Test Accuracy: 0.0084

```
In [24]: # Función para normalizar imágenes solo para visualización
def normalize_for_display(img):
    img_min = img.min()
    img_max = img.max()
    if img_max > 1.0 or img_min < 0.0:
        img = (img - img_min) / (img_max - img_min + 1e-7)
    return img

# Obtener nombres de clases
class_indices = test_generator.class_indices
class_names = list(class_indices.keys())
```

```
# Reiniciar el generador para empezar desde el principio
test_generator.reset()
x_batch, y_true_batch = next(test_generator)

# Obtener predicciones del modelo
y_pred_probs_batch = model.predict(x_batch)
y_pred_batch = np.argmax(y_pred_probs_batch, axis=1)
y_true_batch = np.argmax(y_true_batch, axis=1)

# Número de imágenes a mostrar (máximo el tamaño del batch)
num_samples = x_batch.shape[0]

# Mostrar imágenes con etiquetas
plt.figure(figsize=(15, 8))
for i in range(num_samples):
    plt.subplot(3, 4, i + 1)
    img_to_show = normalize_for_display(x_batch[i])
    plt.imshow(img_to_show)
    plt.axis('off')
    plt.title(
        f"Real: {class_names[y_true_batch[i]]}\nPred: {class_names[y_pred_batch[i]]}
        color='green' if y_true_batch[i] == y_pred_batch[i] else 'red'
    )
plt.tight_layout()
plt.suptitle("Muestreo de Predicciones vs. Etiquetas Reales", fontsize=16, y=0.95)
plt.show()
```

1/1 ————— 1s 948ms/step

```

-----
ValueError                                Traceback (most recent call last)
Cell In[24], line 28
    26 plt.figure(figsize=(15, 8))
    27 for i in range(num_samples):
--> 28     plt.subplot(3, 4, i + 1)
    29     img_to_show = normalize_for_display(x_batch[i])
    30     plt.imshow(img_to_show)

File /opt/anaconda3/lib/python3.12/site-packages/matplotlib/pyplot.py:1534,
in subplot(*args, **kwargs)
    1531 fig = gcf()
    1533 # First, search for an existing subplot with a matching spec.
-> 1534 key = SubplotSpec._from_subplot_args(fig, args)
    1536 for ax in fig.axes:
    1537     # If we found an Axes at the position, we can reuse it if the us
er passed no
    1538     # kwargs or if the Axes class and kwargs are identical.
    1539     if (ax.get_subplotspec() == key
    1540         and (kwargs == {}
    1541             or (ax._projection_init
    1542                 == fig._process_projection_requirements(**kwargs)))
    ):

File /opt/anaconda3/lib/python3.12/site-packages/matplotlib/gridspec.py:589,
in SubplotSpec._from_subplot_args(figure, args)
    587 else:
    588     if not isinstance(num, Integral) or num < 1 or num > rows*cols:
--> 589         raise ValueError(
    590             f"num must be an integer with 1 <= num <= {rows*cols}, "
    591             f"not {num!r}"
    592         )
    593     i = j = num
    594 return gs[i-1:j]

ValueError: num must be an integer with 1 <= num <= 12, not 13

```




```
In [31]: from sklearn.metrics import roc_curve, auc
from tensorflow.keras.utils import to_categorical

# 3. Convertir etiquetas reales a one-hot
n_classes = y_pred_probs.shape[1]
y_true_onehot = to_categorical(y_true, num_classes=n_classes)

# 4. Calcular ROC y AUC por clase
fpr = dict()
tpr = dict()
roc_auc = dict()

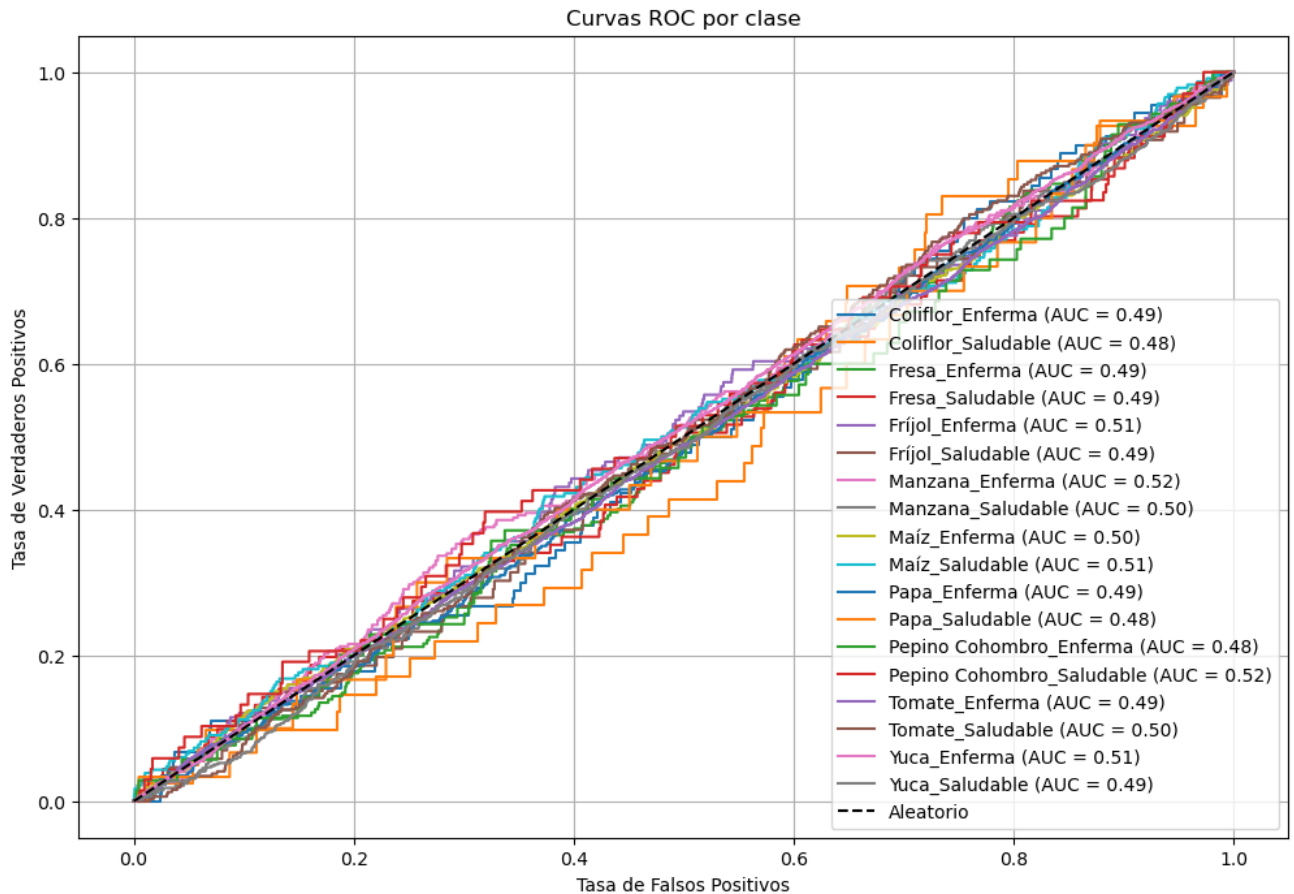
for i in range(n_classes):
    fpr[i], tpr[i], _ = roc_curve(y_true_onehot[:, i], y_pred_probs[:, i])
    roc_auc[i] = auc(fpr[i], tpr[i])

# 5. Obtener nombres de clase
class_names = list(test_generator.class_indices.keys())

# 6. Graficar curvas ROC
plt.figure(figsize=(12, 8))
for i in range(n_classes):
    plt.plot(fpr[i], tpr[i], label=f'{class_names[i]} (AUC = {roc_auc[i]:.2f})')

plt.plot([0, 1], [0, 1], 'k--', label='Aleatorio')
plt.xlabel('Tasa de Falsos Positivos')
```

```
plt.ylabel('Tasa de Verdaderos Positivos')
plt.title('Curvas ROC por clase')
plt.legend(loc='lower right')
plt.grid(True)
plt.show()
```



```
In [33]: from sklearn.metrics import roc_auc_score

# AUC macro (promedio de clases, todas igual de importantes)
auc_macro = roc_auc_score(y_true_onehot, y_pred_probs, average='macro')

# AUC micro (promedia sobre todas las instancias)
auc_micro = roc_auc_score(y_true_onehot, y_pred_probs, average='micro')

print(f"AUC Macro promedio: {auc_macro:.4f}")
print(f"AUC Micro promedio: {auc_micro:.4f}")
```

AUC Macro promedio: 0.4958
AUC Micro promedio: 0.3939

```
In [35]: from sklearn.metrics import f1_score

# Asegúrate de tener esto:
# y_pred = np.argmax(y_pred_probs, axis=1)
# y_true = test_generator.classes
```

```

# Accuracy
accuracy = accuracy_score(y_true, y_pred)

# F1 Scores
f1_macro = f1_score(y_true, y_pred, average='macro')
f1_micro = f1_score(y_true, y_pred, average='micro')
f1_weighted = f1_score(y_true, y_pred, average='weighted')

print(f"Accuracy: {accuracy:.4f}")
print(f"F1 Score (macro): {f1_macro:.4f}")
print(f"F1 Score (micro): {f1_micro:.4f}")
print(f"F1 Score (weighted): {f1_weighted:.4f}")

# Detalle por clase
class_names = list(test_generator.class_indices.keys())
print("\nReporte de clasificación por clase:")
print(classification_report(y_true, y_pred, target_names=class_names))

```

```

Accuracy: 0.0084
F1 Score (macro): 0.0015
F1 Score (micro): 0.0084
F1 Score (weighted): 0.0002

```

Reporte de clasificación por clase:

	precision	recall	f1-score	support
Coliflor_Enferma	0.01	0.98	0.02	90
Coliflor_Saludable	0.00	0.00	0.00	41
Fresa_Enferma	0.00	0.00	0.00	222
Fresa_Saludable	0.00	0.00	0.00	91
Fríjol_Enferma	0.00	0.00	0.00	174
Fríjol_Saludable	0.01	0.01	0.01	86
Manzana_Enferma	0.00	0.00	0.00	306
Manzana_Saludable	0.00	0.00	0.00	329
Maíz_Enferma	0.00	0.00	0.00	538
Maíz_Saludable	0.00	0.00	0.00	232
Papa_Enferma	0.00	0.00	0.00	400
Papa_Saludable	0.00	0.00	0.00	30
Pepino Cohombro_Enferma	0.00	0.00	0.00	70
Pepino Cohombro_Saludable	0.00	0.00	0.00	68
Tomate_Enferma	0.00	0.00	0.00	3314
Tomate_Saludable	0.00	0.00	0.00	318
Yuca_Enferma	0.00	0.00	0.00	3764
Yuca_Saludable	0.00	0.00	0.00	516
accuracy			0.01	10589
macro avg	0.00	0.05	0.00	10589
weighted avg	0.00	0.01	0.00	10589

```
In [37]: model.save('modelo_17_EfficientNet_Planta_Estado.h5')
```

WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save\_model(model)`. This file format is considered legacy. We recommend using instead the native Keras format, e.g. `model.save('my\_model.keras')` or `keras.saving.save\_model(model, 'my\_model.keras')`.

```
In [39]: class_indices = train_generator.class_indices # o test_generator.class_indices
index_to_class = {v: k for k, v in class_indices.items()}
print(index_to_class)
with open("index_to_class_modelo_17_EfficientNet_Planta_Estado.json", "w") as f:
    json.dump(index_to_class, f, indent=4, ensure_ascii=False)
```

```
{0: 'Yuca_Enferma', 1: 'Yuca_Saludable', 2: 'Fresa_Saludable', 3: 'Papa_Enferma', 4: 'Maíz_Saludable', 5: 'Tomate_Enferma', 6: 'Manzana_Enferma', 7: 'Maíz_Enferma', 8: 'Tomate_Saludable', 9: 'Manzana_Saludable', 10: 'Fresa_Enferma', 11: 'Papa_Saludable', 12: 'Coliflor_Enferma', 13: 'Pepino Cohombro_Saludable', 14: 'Fríjol_Enferma', 15: 'Coliflor_Saludable', 16: 'Fríjol_Saludable', 17: 'Pepino Cohombro_Enferma'}
```

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In [ ]:
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